



Physics of Eclipsing Binaries. VI. Hot, Compact Stars

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Abstract

Models of eclipsing binaries require the assignment of appropriate emergent intensities to the surface elements of the binary components. For distance-dependent modeling of flux-calibrated light curves, this necessitates an approximation of the absolute normal intensities of both components of the binary, as well as how their brightness varies across the stellar disks (limb darkening). Such surface intensities are often inferred from other physical properties of the synthetic binary (effective temperature, surface gravity, etc.) through the use of model atmospheres, which in turn are generally suited to a particular range of stellar types or parameters. Here, we present the major developments included in the PHOEBE 2.5 release (publicly available from <http://phoebe-project.org>), which improve the fidelity of model binaries comprising hot, compact stars. These developments include the incorporation of model atmospheres produced using the Tübingen Model-Atmosphere Package and Montréal/Tremblay codes (complementing the already incorporated PHOENIX and Castelli & Kurucz models, primarily suited to main-sequence stars and low-temperature giants). Similarly, PHOEBE v2.5 now allows for blending/extrapolation of model atmospheres, meaning one can continue to make use of model atmospheres in cases when a small number of surface elements have parameters outside the model-atmosphere grid. As an added value product, we also present tables of limb-darkening coefficients derived from the newly incorporated model atmospheres, such that they can be used as inputs in other binary modeling codes.

Unified Astronomy Thesaurus concepts: Binary stars (154); Light curves (918); Spectroscopic binary stars (1557); White dwarf stars (1799)

Materials only available in the online version of record: [tar.gz file](#)

1. Introduction

An essential element in all binary light-curve modeling codes is the assignment of an appropriate emergent intensity to all visible surface elements of the discretized mesh that describes the geometry of the two stars, such that (when integrated) the total light from the binary can be modeled (R. E. Wilson & E. J. Devinney 1971; A. Prša 2018). In some codes, the emergent intensities are derived via a user-provided surface-brightness ratio in the passband of interest, e.g., JKTEBOP (J. Southworth 2013) or ELLC (P. F. L. Maxted 2016), while others (including PHOEBE; A. Prša et al. 2016) make use of model atmospheres. Here, the passband-integrated intensities are derived based on synthetic spectra created using model atmospheres, and then tabulated as a function of the defining model parameters (typically effective temperature, surface gravity, and chemical composition; W. van Hamme & R. E. Wilson 2003). The resulting tables can then be interpolated to derive emergent intensities for a given surface

element based on its model properties—again usually the temperature (accounting for irradiation, gravity brightening, etc.), surface gravity, and metallicity. This approach has the added benefit of allowing the model light curves to be calculated in absolute units (taking into account the distance and interstellar extinction of the system; D. Jones et al. 2020), which can then be directly compared to flux-calibrated light curves. However, the model atmospheres used (as well as the codes used to synthesize spectra from these models) are all subject to their own underlying assumptions and, as such, are generally valid only for a limited parameter space. Outside of these ranges, many codes default to blackbody atmospheres, which are, at best, a poor approximation (A. Prša et al. 2016).

PHOEBE was initially developed to make use of synthetic spectra produced with F. Castelli & R. L. Kurucz (2003, referred to in PHOEBE as CK2004) model atmospheres, which assume local thermodynamic equilibrium (LTE)—an assumption that breaks down for both the hottest and coolest stars. In D. Jones et al. (2020), non-LTE PHOENIX atmospheres were incorporated (v2.2 release), extending the useful range of atmospheres to both lower temperatures and lower metallicities. In this paper, we present the implementation of plane-parallel model atmospheres from the Tübingen non-LTE Model-Atmosphere Package (TMAP; T. Rauch &



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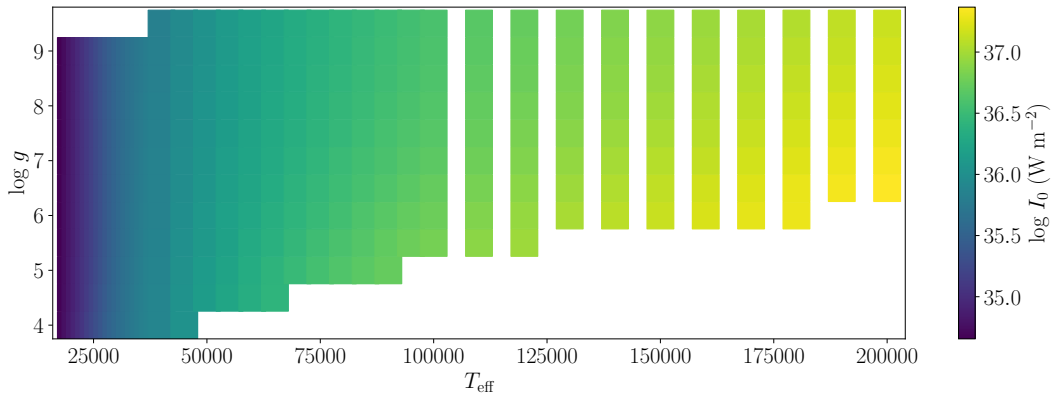


Figure 1. The normal intensity in the SDSS g' band as a function of surface gravity and effective temperature for the TMAP DA WD models. The white spaces are due to the sparser sampling of the grid for higher temperatures and regions where the models approach or exceed the Eddington limit.

Table 1
New Atmosphere Models Implemented into PHOEBE

PHOEBE Name	T_{eff} (kK)	$\log g$ (dex)	$\log(\text{He}/\text{H})$ (dex)	References
tmap_sdO	40–140	4.75–6.5	−1.5– −0.42	N. Reindl et al. (2016)
tmap_DA	20–200	4.0–9.5	−10 ^a	N. Reindl et al. (2023)
tmap_DO	40–200	6.0–9.5	9.4	N. Reindl et al. (2023)
tmap_DAO	40–200	6.0–9.5	−5 – 0	N. Reindl et al. (2023)
tremblay	3.75–60	5.0 – 9.0	−10 ^a	P. E. Tremblay & P. Bergeron (2009)

Notes. Models close to or above the Eddington limit are not included.

^a The implemented atmospheric models are actually pure hydrogen but are labeled as $\log(\text{He}/\text{H}) = -10$ in order to share the same definition of the PHOEBE “loghefrac” parameter between the different hot, compact star models.

J. L. Deetjen 2003; K. Werner et al. 2003) and Montréal/Tremblay group (P. E. Tremblay & P. Bergeron 2009; P. E. Tremblay et al. 2013). These model atmospheres dramatically increase the fidelity of PHOEBE-computed observables for hot and compact stars—including hot white dwarf (WD) and subdwarf O stars. Additionally, we outline several other functionalities implemented in the PHOEBE code as part of the v2.5 release. These include blending (sometimes referred to as ramping; W. van Hamme & R. E. Wilson 2003), Doppler boosting/beamings, and user-defined features. Blending allows for a smooth transition from model atmospheres to blackbodies, in order to avoid the need to fall back on the (demonstrably poor) blackbody approximation for entire stars when only a handful of surface elements lie outside the parameter range covered by model atmospheres (e.g., due to the effects of irradiation or tidal distortion). Doppler boosting was previously disabled in PHOEBE due to concerns over the way boosting factors were derived and tabulated; however, the underlying functionality was never in question. Thus, Doppler boosting is now reactivated, allowing users to provide factors calculated outside of PHOEBE. User-defined features allow the user to modify models after computation or modify component mesh quantities. Example applications of this functionality include differential rotation, migrating spots, and variable third light.

2. Hot, Compact Star Atmospheres

2.1. TMAP Implementation

Angle-dependent specific intensities were calculated using TMAP assuming pure hydrogen or hydrogen–helium atmospheres

and following the four grids (sdO, DA, DAO, DO) presented in N. Reindl et al. (2016, 2023). These grids are implemented as separate atmosphere options inside PHOEBE to avoid any possible issues with interpolation between the different models (although we will later demonstrate that their behavior is consistent). The coverage of the different model grids, in terms of effective temperature, surface gravity, and surface abundances, are detailed in Table 1.

The angles across the stellar limb for which emergent spectra can be calculated are hard-coded into TMAP, and cover 32 values of θ from 2.997 (close to the center of the stellar disk) to 89.922 (close to the edge of the stellar limb). Just as for the other atmospheres incorporated into the code (see A. Prša et al. 2016; D. Jones et al. 2020 and references therein), PHOEBE’s passband files contain tables of these TMAP angle-dependent spectral energy distributions (SEDs) multiplied by the passband transmission function to provide integrated emergent intensities as a function of the surface parameters: effective temperature, surface gravity, and abundance. An example of the variation of normal intensity as a function of surface gravity and effective temperature for DA atmospheres is shown in Figure 1.

The major difference in implementation between the TMAP tables and those of the previously implemented atmospheres is the definition of the abundance (“abun” in PHOEBE). For CK2004 and PHOENIX, the abundance is defined as the metallicity; however, this definition is not appropriate for the hot and compact stars covered by the TMAP grids. As such, to avoid confusion with the “abun” parameter, the abundances of the TMAP grids are specified using a new parameter, “loghefrac,” defined as the logarithmic helium-to-hydrogen number fraction ($\log(\text{He}/\text{H})$).

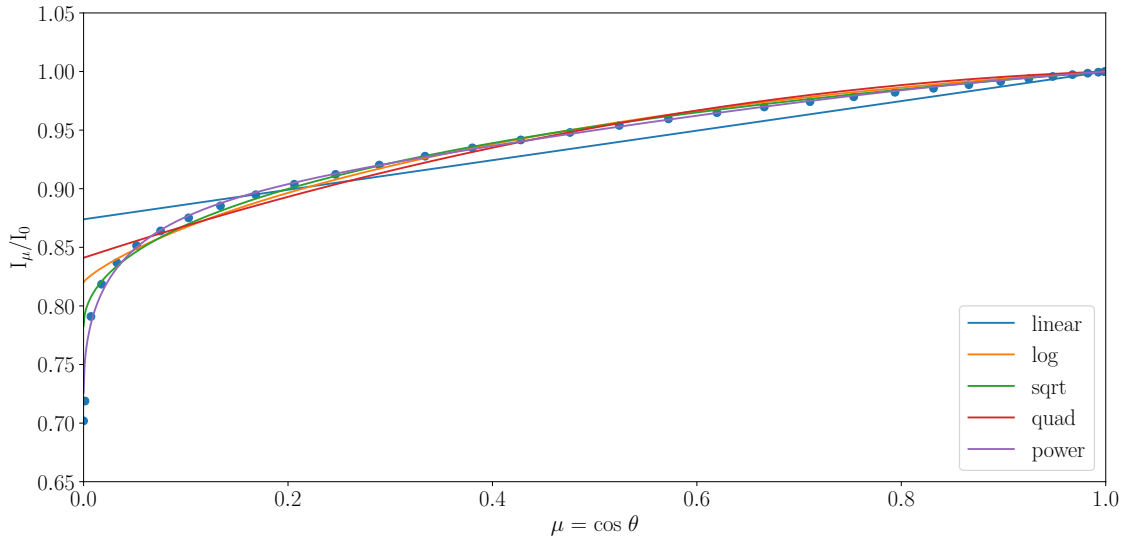


Figure 2. The variation of emergent intensities in the SDSS g band as a function of angle across the stellar disk for a TMAP sdO model with $T_{\text{eff}} = 70$ kK, $\log g = 6.0$, and a $\log(\text{He}/\text{H}) = -0.97$. The blue points show the TMAP sampling along the stellar disk, while the colored lines represent fits to these points using different limb-darkening parameterizations.

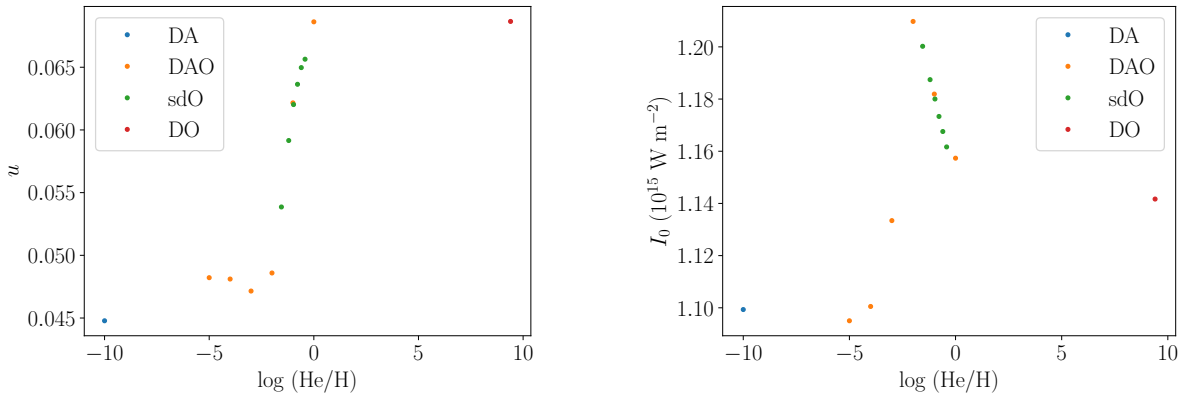


Figure 3. The SDSS g' -band linear limb-darkening coefficient (u ; left) and normal intensity (I_0 ; right) of the four TMAP model grids versus the $\log(\text{He}/\text{H})$ abundance for models with an effective temperature of 80 kK and surface gravity $\log g = 6.5$.

2.2. Limb-darkening Properties of the TMAP Models

An example of the variation in emergent intensity as a function of $\mu = \cos \theta$ for a TMAP atmosphere in the g band is presented in Figure 2. As an added value product, these angle-dependent passband intensities can also be used to derive limb-darkening coefficients for the standard parameterizations used in the literature (see Section 5.2.3 of A. Prša et al. 2016).¹⁰ While they are not required for PHOEBE (which interpolates these tables on the fly at runtime, thus avoiding the need to choose a parameterized approximation; A. Prša et al. 2016), they may be useful for other codes¹¹ and, as such, we provide machine-readable versions of the tables of these coefficients for the standard Sloan Digital Sky Survey (SDSS) and Johnson filter sets for DA, DO, and DAO WDs as well as sdO stars in a .tar.gz package. These values are derived by nonlinear least-squares fitting to the angle-dependent emergent intensities via the Levenberg–Marquardt algorithm (J. J. Moré 1978).

2.3. Consistency of the Four TMAP Grids

The four TMAP model grids provide irregular coverage of the parameter space—particularly in terms of the abundance—however there are overlaps, specifically between the sdO and DAO grids. Thus, we explore the key properties of the models (in terms of binary modeling) in order to test for consistency between the different grids. To do this, we compare the normal intensities and derived linear limb-darkening coefficients as a function of abundance (He-to-H fraction by number) for a given combination of effective temperature and surface gravity in the range covered by all four grids (i.e., $40 \text{ kK} \leq T_{\text{eff}} \leq 140 \text{ kK}$ and $6.0 \leq \log g \leq 6.5$). An example, for $T_{\text{eff}} = 80 \text{ kK}$ and $\log g = 6.5$, is shown in Figure 3, demonstrating the continuous nature of the variation of the limb-darkening coefficient and normal intensity as a function of abundance.

2.4. Montréal/Tremblay Atmospheres

The only other tables of limb-darkening parameters covering a similar range as those of our TMAP models were published by A. Claret et al. (2020a), covering $T_{\text{eff}} \leq 100 \text{ kK}$ and $5.0 \leq \log g \leq 9.5$. They present two overlapping grids for

¹⁰ In A. Prša et al. (2016), there is a typographic error in the definition of the logarithmic model where $\log_{10}$ should, in fact, be the natural logarithm, $\ln$ (D. A. KlingleSmith & S. Sobieski 1970).

¹¹ PHOEBE can also be used to interpolate these tables of limb-darkening coefficients for use in other codes should the user wish.

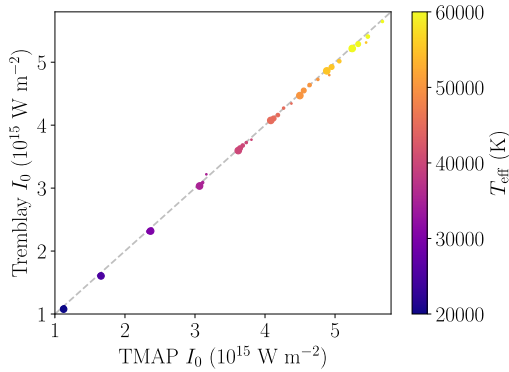


Figure 4. The normal intensities of the TMAP and Tremblay DA models in the SDSS g' band, where the point size is proportional to the surface gravity and the color represents the effective temperature. The dashed line demarcates a perfect agreement between the models.

DA WDs, comprising non-LTE TLUSTY models (I. Hubeny & T. Lanz 1995) extending to higher temperatures and LTE Montréal/Tremblay models reaching down to lower temperatures (P. E. Tremblay & P. Bergeron 2009). The latter models are now also available as an atmosphere option in PHOEBE, with the implementation based upon the same grid of model spectra employed by A. Claret et al. (2020a). For further details of the underlying models, we therefore refer the reader to A. Claret et al. (2020a). We do not incorporate the TLUSTY models of A. Claret et al. (2020a) into PHOEBE as an error in their calculations leads to erroneous intensities and limb-darkening coefficients (P.-E. Tremblay 2026, private communication).¹²

2.5. Comparison of TMAP and Tremblay Limb-darkening and Normal Intensities

As a first step, we check the consistency between the TMAP DA and Tremblay models as implemented in PHOEBE, contrasting the normal intensities in the SDSS g' passband of both models as a function of effective temperature and surface gravity in Figure 4. The agreement between the two models is clear, with the ratio of the normal intensities being within 1 standard deviation of unity (1.01 ± 0.10 for SDSS g').

A. Claret et al. (2020a) provide the normal intensities for their models as $\text{mW m}^{-2} \text{Hz}^{-1}$ requiring a passband- (and thus spectrum-) dependent conversion to W m^{-2} for direct comparison with the models implemented in PHOEBE.¹³ For this, we assume that the effective wavelength of the passband is also the effective wavelength of the passband convolved with the emergent spectrum (i.e., the emergent spectrum is essentially flat in the range covered by the passband). Under this assumption, the agreement between the Tremblay intensities tabulated by A. Claret et al. (2020a) and the intensities in PHOEBE are good—again being within 1 standard deviation of unity.

¹² It is important to emphasize that this is not an error in TLUSTY or reflective of all synthetic spectra calculated using TLUSTY. Extensive testing and comparison between TMAP and TLUSTY has found that the two codes agree to within 1% over an extended wavelength range (R. C. Bohlin et al. 2020). In this case, it is simply an error in the intensities presented by A. Claret et al. (2020a).

¹³ According to the notes added to the VizieR catalogs, the A. Claret et al. (2020a) tables also require the application of an additional factor of 2, which are taken into account in the PHOEBE implementation of the Tremblay models.

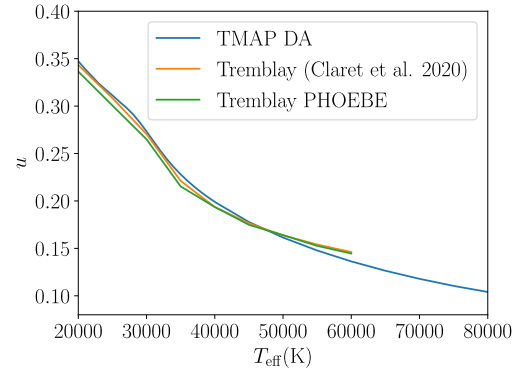


Figure 5. The linear limb-darkening coefficient, u , as a function of effective temperature for DA models with $\log g = 7.0$.

Comparing limb-darkening coefficients leads to a very similar conclusion. There is generally very good agreement between the values measured by A. Claret et al. (2020a) from the Tremblay models and the values from PHOEBE using the same models as well as using TMAP (see Figure 5). Small deviations are found between the Tremblay coefficients from A. Claret et al. (2020a) and PHOEBE, principally due to the methodology used to fit the models. A. Claret et al. (2020a) interpolate the models to 100 equally spaced points along the stellar limb, while PHOEBE takes the 20 unevenly spaced points and weights them by the size of the intervals between points. The agreement between the Tremblay and TMAP models is similar to that found when comparing normal intensities.

3. TMAP/Tremblay versus Blackbody Atmospheres

It has been clear for some time that approximating hot, compact stars as blackbodies can lead to significant issues, for example when considering the central ionizing sources of photoionized nebulae (e.g., S. R. Pottasch et al. 2008). Here, we attempt to characterize the impact that this can have on a modeled binary system.

Across the optical, TMAP passband-integrated intensities are approximately 25% lower than those of a blackbody with the same temperature (see Figure 6; e.g., T. Rauch 2003), depending on the passband, effective temperature, surface gravity, and abundance (in some instances, the difference is as large as 40%). The principal impact is that, for a given passband, the temperature of a hot, compact star is likely to be underestimated when approximated by a blackbody in a model light curve.

When such hot and compact stars, like those covered by the TMAP model grids, are found in close binaries with lower-temperature companions, they are often found to irradiate the face of the companion closest to them, leading to a roughly sinusoidal variation as a function of orbital phase—referred to as an irradiation or reflection effect (M. Horvat et al. 2019). The shape and amplitude of this effect is a complex function of the passband and the binary parameters (e.g., S. H. P. Alencar & L. P. R. Vaz 1999; D. Jones & H. M. J. Boffin 2017), but perhaps most critically, at least in the context of the comparison between TMAP and blackbody atmospheres, upon the luminosity of the irradiating star and the bolometric albedo of the irradiated star. As mentioned above, the difference in optical intensities between TMAP and blackbody atmospheres will often lead to underestimated temperatures for hot stars when

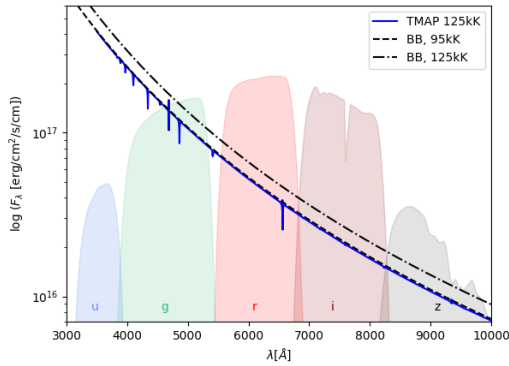


Figure 6. A comparison of a 125 kK TMAP spectrum with blackbody spectra of different temperatures. Standard SDSS filter bandpasses are underplotted to highlight that the differences are appreciable throughout the optical range.

modeling optical light curves. This would also imply a reduced luminosity and thus reduce the level of irradiation experienced by a close companion. Therefore, in order to reproduce the amplitude of an irradiation effect using blackbody atmospheres for the irradiating star, the bolometric albedo of the irradiated companion will need to be increased—in many cases leading to highly unphysical albedos well in excess of unity.

To demonstrate the issues highlighted above, we create a synthetic binary comprising a young, 127 kK pre-WD (with mass, temperature, and radius taken from the post-asymptotic giant branch evolutionary tracks of M. M. Miller Bertolami 2016) in an 8 hr orbit with a K5V companion. Model light curves of this synthetic binary in the SDSS *i* band, calculated using TMAP and blackbody atmospheres, are shown in Figure 7, clearly demonstrating that the blackbody atmosphere results in a brighter pre-WD (as evidenced by the model light curve being ~ 0.3 mag brighter upon egress of the primary eclipse). In order to roughly reproduce the TMAP model light curve with blackbody atmospheres, the temperature of the pre-WD needs to be reduced to approximately 94 kK ($\sim 74\%$ the original T_{eff} used in the TMAP model) in the blackbody model. Similarly, in order to compensate the reduced luminosity of the pre-WD with its reduced effective temperature in the new blackbody model, the albedo of the companion needs to be significantly increased (from 0.6 to an unphysical 2.0!) in order to reproduce the amplitude of the irradiation effect.

4. Blending

New model atmospheres also highlight another problem that PHOEBE has long struggled with: the functionality when atmospheric parameters go out of the range covered by the model atmospheres. There are several common cases in which this can become an issue, for example:

1. a semidetached binary in which a few surface elements near the inner Lagrange point become too cool or too rarefied for model atmospheres;
2. a hot, compact star that irradiates its companion, thereby exceeding model-atmosphere boundaries;
3. a rapidly rotating star whose equatorial surface gravity falls outside the grid.

In these and similar circumstances, PHOEBE previously raised an error that the chosen model atmosphere could not be used. The question is whether this is indeed warranted: Should the benefits of the sophisticated model atmosphere be lost on

account of a few surface elements that might fall outside the model-atmosphere grid?

To overcome this limitation, PHOEBE now supports two modes of operation that allow for a more nuanced treatment of model atmospheres: *extrapolation* and *blending*. We discuss both, along with the typical circumstances that warrant their use, below. The underlying principle is the construction of an incompletely populated n -dimensional (n -D) grid of physical quantities (intensities, limb-darkening coefficients, boosting coefficients, etc.) that stem from model atmospheres.

Consider a model atmosphere with three basic input parameters: effective temperature (T_{eff}), surface gravity ($\log g$), and heavy metal abundance ($[M/H]$). These parameters determine the basic axes of a three-dimensional space. In principle, given the values of the three parameters and assuming that they correspond to a physically allowed circumstance, we could use a model-atmosphere code to synthesize a SED, multiply it by the passband response function, and ultimately integrate it into an emergent passband intensity. This is computationally prohibitive given that there are as many combinations of atmospheric parameters as there are surface elements on the star. Instead, each model atmosphere is sampled in discrete combinations of T_{eff} , $\log g$, and $[M/H]$, the corresponding SEDs precomputed, and emergent passband intensities tabulated in a grid. As not all combinations of T_{eff} , $\log g$, and $[M/H]$ are physical, some grid nodes will remain undefined. Hence, we have constructed an incompletely populated n -D grid of emergent intensities that PHOEBE can then use to linearly interpolate at any given three-dimensional point (T_{eff} , $\log g$, $[M/H]$).

Linear interpolation on an incompletely populated grid is not a well-posed problem. To demonstrate this, consider a two-dimensional case with axes $a = (1, 2, 3)$ and $b = (4, 5, 6)$. This 3×3 grid requires nine function values to be fully determined. Now suppose that a function value at $(2, 6)$ is unphysical or otherwise unavailable. How would interpolation be done in $(2.3, 5.3)$? We clearly have an overdimensioned problem in which any four nodes among the defined function values can be used. Given that function values (for example, intensities) are generally nonlinear in basic parameters, the decision of what nodes to use for interpolation *matters*. We elect to use the nearest *fully defined hypercube*, in this case $(2, 3) \times (4, 5)$. The rationale, along with a more formal discussion of the interpolation strategy and the used terminology, is given in the [Appendix](#).

The same construct can be used for linear extrapolation. This, too, is not a well-posed problem. Extrapolating to an off-grid value again requires decisions on what defined nodes to use, which again matters because of the nonlinearity of the function values. We discuss these considerations in the [Appendix](#) as well.

Here, we will use a catch-all term, “ndpolant,” to stand for either interpolant or extrapolant in n -D space, and “ndpolation” as the generalized term for linear interpolation and extrapolation in n -D space. Pertinent to the discussion at hand is that *there exists* an ndpolant that can be computed from an incomplete n -D grid.

Suppose that we want to evaluate a function value in an out-of-bounds query point $q = (T_{\text{eff}}, \log g, [M/H])$. PHOEBE supports two modes of operation:

1. *Extrapolation*. Driven by the PHOEBE parameter `extrapolation method`, which can be set to “none”, “nearest”, or “linear”. If “none”, extrapolation will

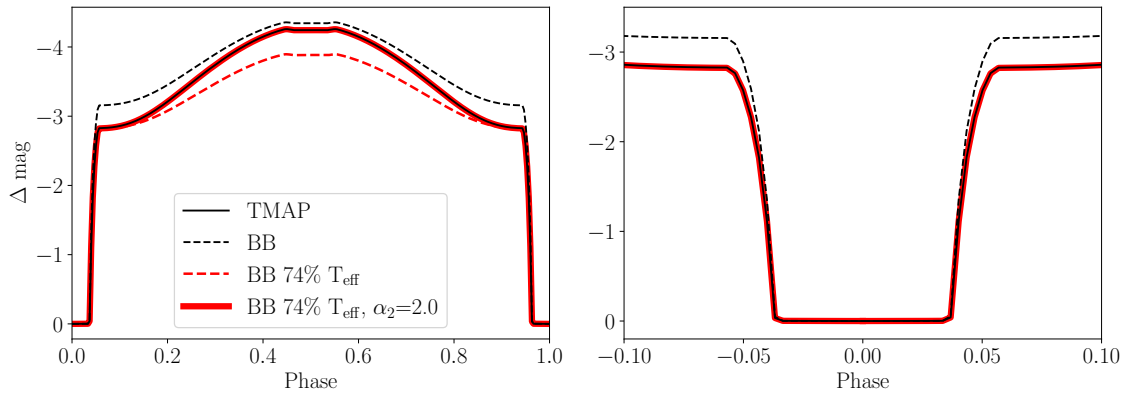


Figure 7. A synthetic binary comprising a pre-WD ($T_{\text{eff}} = 127$ kK with TMAP atmospheres) in an 8 hr orbit with a K5V companion (TMAP; solid black curve). For comparison, the same model was run with a blackbody atmosphere for the pre-WD (BB; dashed black line), with a blackbody atmosphere and a reduced temperature ($T_{\text{eff}} \sim 94$ kK, labeled “BB 74% T_{eff} ”; dashed red line) and with a blackbody atmosphere of reduced temperature and increased secondary albedo (labeled “BB 74% T_{eff} , $\alpha_2 = 2.0$ ”; solid red line).

not be used; this corresponds to the pre-2.5 functionality. The code will raise an error and stop execution. If “nearest”, the nearest defined node will be used, where nearest is defined by the renormalized Euclidean distance (see the [Appendix](#)). This is most appropriate for extrapolating limb-darkening coefficients that are expressly nonlinear. Finally, if “linear,” the function value will be linearly extrapolated from the nearest fully defined hypercube. This is most appropriate for extrapolating normal and specific emergent intensities.

2. *Blending.* Driven by the PHOEBE parameter `blending_method`, which can be set to “none” or “blackbody”. If “none,” the code will raise an error and stop execution. If “blackbody,” the function value will be a blend between the extrapolated value from the given model atmosphere and the blackbody value, with the blending ratio dependent on the distance from the model-atmosphere grid. This achieves a gradual transition from on-grid values to extrapolated values in the immediate vicinity to the grid that progressively morph into blackbody values as the distance from the grid increases. Blending is only appropriate for the normal emergent intensities as that is the only quantity inherent to the blackbody model atmosphere.

With this new functionality, PHOEBE can now compute observables even if some (or all) surface elements fall outside the grid of supported model atmospheres, by either extrapolating or blending into blackbody atmospheres. Each such extrapolation or blending is stored in the mesh and presented to the user. This significantly facilitates the use of PHOEBE on the less well-populated parts of the H-R diagram, as well as in systems with significant tidal and/or rotational distortion.

5. Boosting

Support for relativistic boosting (also frequently referred to as Doppler boosting or Doppler beaming) has been disabled in PHOEBE since the v2.2 release due to concerns over the way the boosting factors were derived from model atmospheres.¹⁴ However, as of v2.5 we return this functionality, allowing the user to provide the boosting factors—either taking them from

the literature (e.g., A. Claret et al. 2020b) or fitting for them. This approach has already been successfully used, for example, by J. Munday et al. (2023).

6. User-defined Features

This release of PHOEBE also introduces the new capability to define custom code that can be executed at several different steps within PHOEBE’s logic:

1. Modifying the mesh before local value computations at any given model time.
2. Modifying local mesh values at any given model time: radial velocities, surface gravities, effective temperatures, and passband intensities.
3. Modifying the completed synthetic model prior to evaluating the merit function.

In order to access this functionality, the user writes a class that inherits from a provided class in PHOEBE, defines the input parameters, and optionally overrides any number of these “hooks” into PHOEBE. This class is then used to add a “feature” to the PHOEBE system, in the same way as previously built-in features including spots. The parameterization and code logic are then serialized along with the PHOEBE object (or saved file) and can be moved to another machine without having to move additional code or install extra dependencies.

This design allows any custom code to be portable (including making use of external servers to offload expensive computations), shareable, and compatible with distributions and solver logic introduced in the previous release. Any parameter written by the user is treated the same as a native parameter, enabling them to be optimized or sampled over simultaneously.

Example use cases for these user-defined features might include custom spot shapes and/or migration in time, differential rotation, simple intensity modulation from pulsations, or an additional time-dependent light contribution.

Figure 8, for example, shows a PHOEBE model with a spot that migrates in colatitude, having a more profound effect on the light curve as it approaches the orbital plane/line of sight. In this case, the user writes a class which inherits from the default `Spot` class, defines the parameterization of the time dependence (in this case, the rate at which the spot’s colatitude changes), and overloads the instantaneous position method that

¹⁴ For a complete discussion of the issue, please see http://phoebe-project.org/docs/latest/tutorials/beaming_boosting.

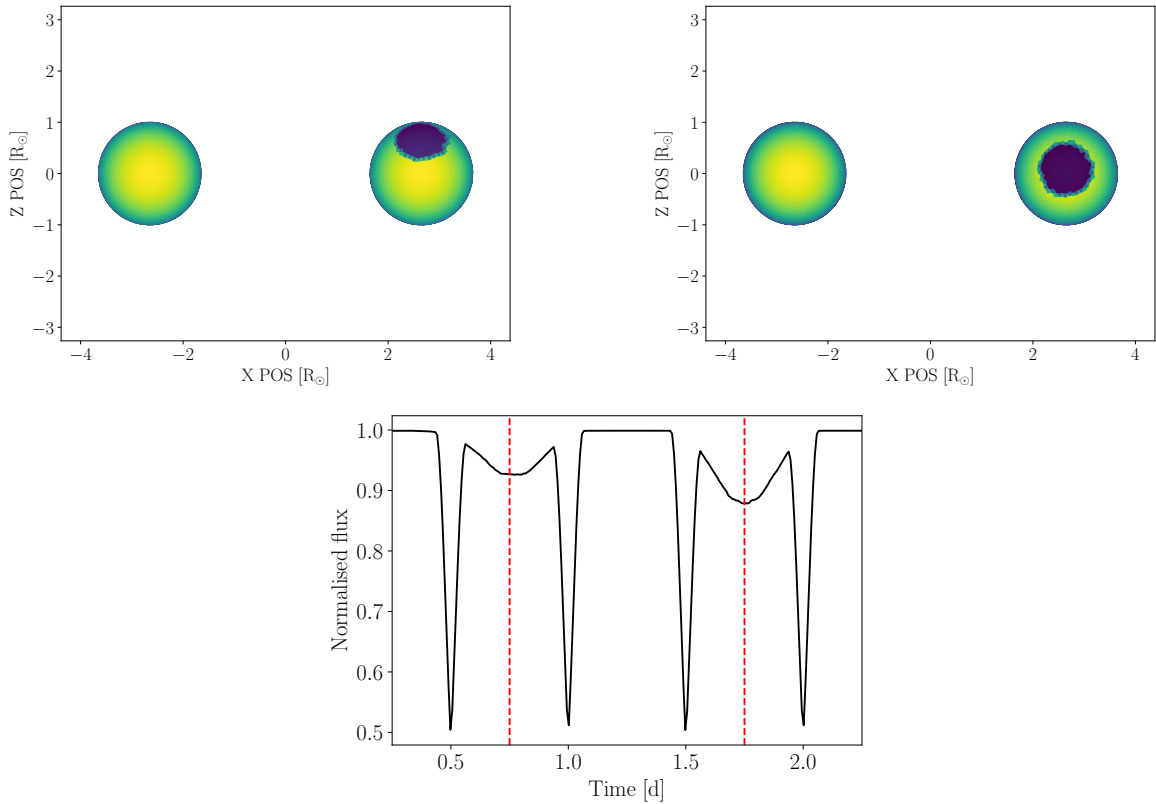


Figure 8. A synthetic binary with a custom user-defined migrating spot. The lower panel shows the synthetic light curve, in which the vertical dashed lines mark the times of the mesh plots shown in the upper panel. Both meshes are at the same orbital phase, but the right-hand panel is one orbit later, when the spot is closer to the line of sight, leading to a more pronounced change in flux as shown in the light curve.

tells the existing spot code which triangles need their temperatures modified at any given time. The code for this example and others can be found in the online PHOEBE documentation for the v2.5 release.

7. Summary

As of the v2.5 release, PHOEBE includes improved support for hot, compact stars via the inclusion of TMAP model atmospheres as well as blending between blackbody and TMAP/Tremblay/PHOENIX/CK2004 atmospheres. This means that it is no longer necessary to use the blackbody approximation for stars whose surface elements have parameters that lie close to those covered by the incorporated model-atmosphere grids. Both of these improvements can have a dramatic impact on the stellar parameters derived by modeling light curves, as evidenced using the TMAP atmospheres, where modeling with a blackbody can underestimate the true effective temperature by as much as $\sim 25\%$. Support for user-provided Doppler-boosting coefficients has also been implemented (following the decision to deprecate interpolated boosting in PHOEBE’s v2.2 release), returning important functionality for cases where this effect can be appreciable (i.e., compact binaries). New capabilities have also been implemented that allow user-defined features, which can modify the PHOEBE mesh, local values of the mesh, or the completed synthetic model.

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Software: Astropy (Astropy Collaboration et al. [2013](#), [2018](#), [2022](#)), Matplotlib (J. D. Hunter [2007](#)), NumPy (T. E. Oliphant [2006](#)), PHOEBE (A. Prša et al. [2016](#); M. Horvat et al. [2018](#); D. Jones et al. [2020](#); K. E. Conroy et al. [2020](#)), SciPy (P. Virtanen et al. [2020](#)), TMAP (T. Rauch & J. L. Deetjen [2003](#); K. Werner et al. [2003](#)).

Appendix

N-dimensional Interpolation on Sparse Grids

Multivariate (n -D) interpolation and extrapolation are techniques used in mathematics, statistics, and science to estimate unknown values between and beyond existing data

points in a multidimensional space. Interpolation involves estimating the function value at points within the range determined by the existing data points, while extrapolation involves estimating the function value beyond that range. There are numerous robust implementations of multivariate interpolation, including k -nearest neighbors (T. Cover & P. Hart 1967), natural neighbor interpolation (R. Sibson 1981), radial basis functions (R. L. Hardy 1971), kriging (N. Cressie 1990), and many others. SciPy, for example, features an entire module for interpolation (`scipy.interpolate`) that implements several multivariate interpolation classes, including piecewise-linear, nearest neighbor, and radial basis function interpolators. Unfortunately, none of the implemented SciPy methods lend themselves readily to extrapolation; at most, they can fill the values off the convex hull with NaNs or a value supplied by the user. In addition, interpolators that operate on a regular n -D grid do not allow any missing data; those values either need to be imputed by using unstructured data interpolators, or structured data interpolators need to be abandoned altogether.

Ndpolator¹⁵ aims to fill this gap: It can both interpolate and extrapolate function values within and beyond the grid definition range, and it can operate on incomplete grids. As a side benefit, ndpolator can estimate both scalar and vector function values, and it can reduce grid dimensionality for points of interest that lie on grid axes. It is optimized for speed and portability (the backend is written in C), and it also features a Python wrapper. Given the gap in the multivariate interpolation and extrapolation landscape, the development of ndpolator has been separated from PHOEBE and made available to the community as a standalone package.

A.1. Ndpolator's Operation Principles

Consider a scalar or a vector field $\mathbf{F}$ that is sampled in a set of n -D points, $\mathbf{F}(x_1, \dots, x_d)_k$, $k = 1 \dots N$. Let these function values be sampled on a grid, where axes $\mathbf{a}_k$ span each grid dimension, so that $\mathbf{X}_{k=1}^d \mathbf{a}_k \equiv \mathbf{a}_1 \times \mathbf{a}_2 \times \dots \times \mathbf{a}_d$ is a Cartesian product that spans the grid. Axis spacing need not be uniform: Vertices can be separated by any amount that is required to make the grid sufficiently locally linear. If the grid is complete, i.e., if there is a function value $\mathbf{F}(x_1, \dots, x_d)$ associated with each grid point, we have $N_c = \prod_k l(\mathbf{a}_k)$ function value samples, where $l(\mathbf{a})$ is the length of axis $\mathbf{a}$. If the grid is incomplete, i.e., if some function values are missing, then $N < N_c$. Ndpolator defines grid points with sampled values as *nodes*, and grid points with missing values as *voids*. The points in which we want to estimate the function value are called *query points* or *points of interest*. The smallest d -dimensional subgrid that encloses (or is adjacent to) the query point is called a *hypercube*.

A.2. Unit Hypercube Transformation

The first, most fundamental principle of ndpolator is that all interpolation and extrapolation is done on *unit hypercubes*. In real-world applications, it is rarely true that all axes are defined on a unit interval. This can lead to vertices of significantly different orders of magnitude along individual axes. To that end, ndpolator first normalizes the hypercubes by transforming them to unit hypercubes: Given the sets of two consecutive

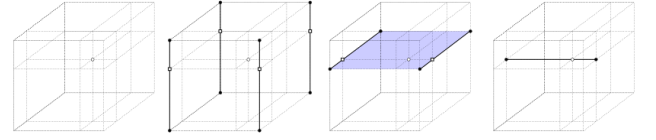


Figure 9. An example of sequential dimensionality reduction in three dimensions.

axis values that span a hypercube, $(\mathbf{a}_{1,p}, \mathbf{a}_{1,p+1}) \times (\mathbf{a}_{2,q}, \mathbf{a}_{2,q+1}) \times \dots \times (\mathbf{a}_{d,t}, \mathbf{a}_{d,t+1})$, the unit transformation maps it to the $[0, 1] \times [0, 1] \times \dots \times [0, 1] \equiv [0, 1]^d$ hypercube. All query points are subjected to the same transformation, creating unit-normalized query points. Therefore, interpolation (and extrapolation) *always* operates on unit hypercubes, which is computationally the least expensive and numerically the most stable process. An additional benefit of this transformation is that extrapolation inherits the nearest hypercube's grid spacing, thus naturally accounting for (potentially) variable spacing in different regions of the grid.

A.3. Sequential Dimensionality Reduction

The second operating principle of ndpolator is *sequential dimensionality reduction*. Consider a three-dimensional hypercube in Figure 9; let us assume that function values in all eight corners of the hypercube are sampled, i.e., we have eight nodes. The point of interest is depicted with an open symbol in the left panel, along with projections onto the hypercube faces. Ndpolator starts with the last axis, in this case $\mathbf{a}_3$, and it interpolates function values along that axis to the projections of the point of interest (second panel). These are *univariate* interpolations. The process yields four vertices (depicted in open symbols), thereby reducing the initial dimension $d = 3$ by 1, to $d - 1 = 2$. The process is then repeated (third panel), this time along the second axis, $\mathbf{a}_2$, yielding two vertices, thereby reducing the dimension to 1. Finally, the last interpolation is done along axis $\mathbf{a}_1$ (right panel), yielding a single vertex, the point of interest itself. The dimension is thus reduced to 0, and the function value is determined. Thus, for an d -dimensional hypercube, ndpolator performs $\sum_{k=0}^{d-1} 2^k = 2^d - 1$ univariate interpolations to estimate $\mathbf{F}$ in the point of interest, which implies an $O(N)$ time complexity, $N = 2^d$ being the number of initial vertices within which one interpolates. This cost is acceptable for d less than roughly 6–7. We note that if the interpolations along each of the dimensions are linear, the resulting value does not depend on the order in which the reductions were performed.

A.4. Initial Dimensionality Reduction

The third operating principle of ndpolator is *initial dimensionality reduction*. In real-life applications, it frequently happens that some query point coordinates are aligned with the axes. For example, one of the axes might allow the variation of the second-order variable, but its value usually defaults to the value that is sampled across the grid. When this happens, the initial hypercube dimension can be reduced by 1 for each aligned axis. The extreme case where the query point coincides with a node means that hypercube dimensionality is reduced to 0, and there is no need for interpolation. For that reason, ndpolator flags each coordinate of the query point as “on-grid,” “on-vertex,” or “out-of-bounds.” When on-vertex, hypercube dimension can be immediately reduced. When that happens,

¹⁵ <https://github.com/aprsa/ndpolator>

the time dependence is reduced by a factor of 2^m , where m is the number of coordinates aligned with the axes.

A.5. Incomplete Hypercubes

The fourth operating principle of ndpolator is dealing with *incomplete hypercubes*. If any of the hypercube corners are voids, we cannot interpolate. For that purpose, ndpolator keeps track of all fully defined n -D hypercubes; when a query point lies within an incomplete hypercube, ndpolator finds the nearest fully defined hypercube and uses it to extrapolate the function value in the point of interest. While this is globally still considered interpolation as the query point is within the grid's definition range, the estimated function value is, strictly speaking, *extrapolated* from the nearest fully defined hypercube. Note that, when grids are particularly sparse and functions strongly nonlinear, that can cause a substantial accumulation of error. In such cases, unstructured interpolation techniques might be a better fit.

A.6. Extrapolation Modes

The fifth operating principle of ndpolator is extrapolation. Ndpolator has three extrapolation methods: “none,” “nearest,” and “linear.” When the extrapolation method is set to “none”, the function value that is outside the range of axes is set to NaN. For the extrapolation method “nearest,” ndpolator stores a list of all nodes and assigns a function value in the node that is nearest to the query point. Lastly, if the extrapolation method is set to “linear,” ndpolator linearly extrapolates from the nearest fully defined hypercube in a manner equivalent to dealing with incomplete hypercubes. The choice of extrapolation method depends on the multivariate function that we are estimating; if it is highly nonlinear, extrapolation should be avoided, so “none” and “nearest” might be appropriate; if it is largely linear or varies slowly, then a “linear” extrapolation method might be warranted. Ndpolator is a linear extrapolator, so it cannot adequately estimate nonlinear multivariate functions.

A.7. Basic Axes and Associated Axes

The question of grid completeness is quite impactful for performance; this is why the sixth operating principle of ndpolator is to distinguish between *basic* axes and *associated* axes. Axes that can have voids in their Cartesian products are referred to as *basic*. For these axes, we need the full ndpolator machinery to perform interpolation and extrapolation. On the other hand, a subset of axes may have all nodes in their Cartesian products, i.e., they are guaranteed to be sampled in all vertices that basic axes are sampled in; these are referred to as *associated* axes. Given that their sampling is ascertained, interpolation and extrapolation can proceed without concerns for incomplete hypercubes—that is, for as long as their basic hypercube counterparts (hypercubes spanned by basic axes) are complete. Each associated axis reduces the dimensionality of the hypercubes that need to be stored for extrapolation lookups, thus optimizing performance further.

A.8. Function Value Dimensionality

The seventh and final operating principle concerns *function value dimensionality*. Most interpolators assume that the function value F is a scalar; ndpolator does not make that assumption. $F_r(x_1, \dots, x_d)$ can be a scalar or a vector or arbitrary length R (within reason, of course). It is then a

requirement that all nodes are also R -dimensional. Ndpolator will then interpolate and extrapolate all function value components separately, and yield an R -dimensional estimate of the function value F in the point of interest.

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